	DIRECT	NATIVE-COT	EN-COT	TRANSLATE-EN
NATIVE-EXEMPLARS	✓	✓	✓	✓
ENGLISH-EXEMPLARS	✓	N/A	✓	N/A
MULTILINGUAL-EXEMPLARS	✓	✓	✓	N/A

Table 2: Possible combinations between few-shot exemplar selection and solution strategies.

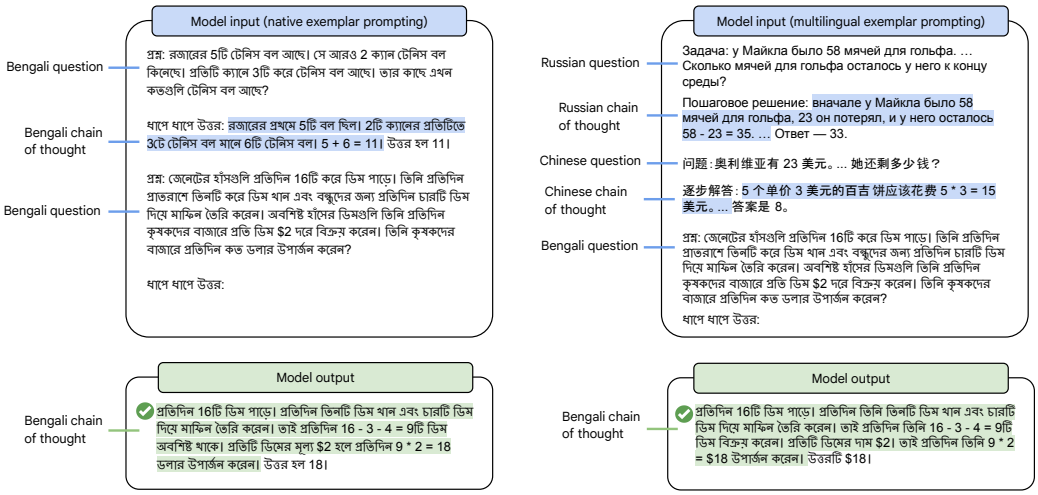

Figure 3: The chain-of-thought prompts and example model outputs in the MGSM experiments. The solutions are written in the same language as the questions of interest (NATIVE-COT).

Beyond the prompting methods, there are different ways to provide few-shot examples in context for multilingual prompting:

- **All native question exemplars (NATIVE-EXEMPLARS).** We use a few in-language questions together with their solutions as the few-shot prompt exemplars. This is the most natural setting when we have a few examples in each investigated language.
- **All English question exemplars (ENGLISH-EXEMPLARS).** When we are unable to access any existing questions or solution examples in some languages, an intuitive way is to use English questions and solutions as exemplars to perform zero-shot cross-lingual transfer. Note that it is unrealistic to combine this exemplar selection setting with NATIVE-COT, since we assume no access to the native language for prompting.
- **Generic multilingual question exemplars (MULTILINGUAL-EXEMPLARS).** Similar to ENGLISH-EXEMPLARS, we assume access to questions and solutions in a few languages, and test if multilingual exemplars better elicit the multilingual reasoning ability of models.

For TRANSLATE-EN, as all exemplar questions and solutions are in English, we only experiment with the translated native question exemplars and English CoT. We summarize the combinations of prompting and exemplar methods in Table 2, and present an illustration in Figure 3. Detailed prompting input for each investigated combination can be found in Appendix A.2.

4 EXPERIMENTS ON MGSM

In this section, we evaluate the multilingual reasoning abilities of two representative state-of-the-art pretrained large language models—GPT-3 (Brown et al., 2020) and PaLM (Chowdhery et al., 2022) —on our MGSM benchmark in various prompting settings using exemplars in the source language

	AVG	HRL	URL	EN	DE	FR	ES	RU	ZH	JA	TH	TE	BN	SW
Lang. Freq. (PaLM, %)	–	–	–	78.0	3.5	3.3	2.1	.53	.40	.38	.04	.02	.006	.005
GPT-3 (`text-davinci-002`)														
• DIRECT	11.7	15.1	5.7	16.0	14.8	16.8	17.2	12.4	18.0	11.2	8.8	0.8	4.4	8.8
• NATIVE-COT	26.4	34.7	7.2	53.6	36.0	37.6	40.4	28.4	40.0	26.0	10.8	0.4	6.4	11.2
• EN-COT	31.6	39.4	13.9	53.6	44.0	46.0	44.8	28.4	40.8	32.4	19.6	5.6	9.6	20.8
• TRANSLATE-EN	45.6	47.5	40.7	53.6	46.4	46.4	51.6	48.8	47.2	44.8	41.2	42.8	41.2	37.6
PaLM-540B														
• DIRECT	18.6	19.3	16.8	22.0	18.8	19.6	20.0	22.0	19.2	16.0	16.8	17.6	17.2	15.6
• NATIVE-COT	48.1	47.9	44.9	**62.4**	49.2	46.4	56.8	48.4	46.8	40.0	52.8	45.6	46.0	35.2
• EN-COT	51.3	52.3	46.8	**62.4**	53.6	51.2	58.0	55.6	46.0	49.6	49.6	46.8	46.4	44.4
• TRANSLATE-EN	**55.0**	**56.3**	**51.2**	**62.4**	**57.2**	**55.2**	**60.0**	**59.6**	**55.6**	**50.0**	**50.8**	**49.6**	**53.2**	**51.2**

Table 3: Accuracy (%) on MGSM of different models and languages with exemplar questions in native languages (NATIVE-EXEMPLARS). HRL: average performance across high-resource languages with larger than 0.1% frequency in the training corpora; URL: average performance across underrepresented languages. We use 6 questions and solutions as the few-shot exemplar whenever possible: while the token number for 6-shot prompts in some languages may exceed the token number limit of GPT-3, we use the maximum possible number of exemplars instead for these cases. Detailed numbers of exemplars for each language in GPT-3 experiments can be found in Appendix A.1. The best numbers in each column are in **boldface**.

(NATIVE-EXEMPLARS).[2] Throughout this paper, we generate outputs using greedy decoding (i.e., sampling with temperature $\tau = 0$).

4.1 MAIN RESULTS

We first compare the few-shot NATIVE-EXEMPLARS performance with different solution strategies (Table 3). In line with the English results reported by Wei et al. (2022b), we find that intermediate reasoning steps (NATIVE-COT and EN-COT) help both models achieve substantial reasoning performance gains across all languages, outperforming direct answer prediction with no explicit reasoning steps (DIRECT) by a significant margin. PaLM shows exceptional multilingual reasoning ability: while it outperforms GPT-3 on all languages with different settings, PaLM-540B with intermediate reasoning steps (NATIVE-COT and EN-COT) achieves results similar to TRANSLATE-EN on all languages, even on underrepresented languages such as Bengali (BN) and Swahili (SW), which cover less than 0.01% of the training corpora.

In addition, reasoning in English (EN-COT) consistently achieves competitive or better performance than reasoning in the native language of the question (NATIVE-COT), suggesting that English intermediate steps can be considered as useful baseline in future work on multilingual reasoning.

4.2 FURTHER ANALYSIS

Effect of language frequency in training corpora. We illustrate the main results of NATIVE-COT, EN-COT and TRANSLATE-EN with respect to the language frequency in PaLM training data (Figure 1). Surprisingly, there is no strong correlation between the performance and the language frequency in the training corpora: the average accuracy among the four underrepresented languages was only 3% lower than the that among the six high-resource languages (44.9% vs 47.9%). Moreover, the performance of reasoning in Thai, Telugu, and Bengali is on par with reasoning in French, Japanese, and Chinese, despite having significantly much less data in the training corpora.

In contrast to prior work that identifies language frequency as important for complex NLU tasks with relatively smaller models (Hu et al., 2020; Lauscher et al., 2020; Ahuja et al., 2022), these results thus indicate that the reasoning ability of large language models may not be primarily dependent on

[2]We focus on these two models due to their notable few-shot performance. In contrast, current multilingual models perform poorly in few-shot settings and are generally used for finetuning with more data (Winata et al., 2021).

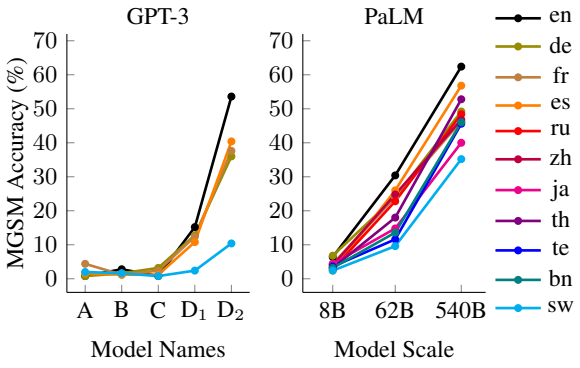

Figure 4: MGSM accuracy with different model scales. The letters A, B, C, D_1, and D_2 denote `text-ada-001`, `text-babbage-001`, `text-curie-001`, `text-davinci-001`, and `text-davinci-002` in the GPT-3 (Brown et al., 2020; Ouyang et al., 2022) family, respectively. While the number of parameters in each GPT-3 model is not publicly available, we order them alphabetically. Detailed numbers can be found in Table 8.

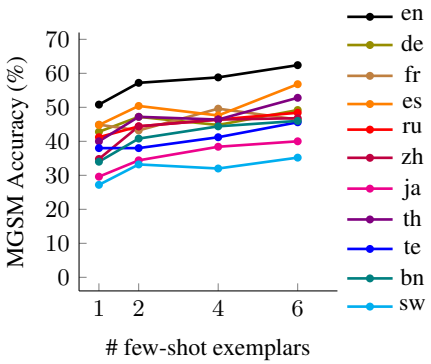

Figure 5: MGSM accuracy of PaLM-540B with different numbers of few-shot exemplars. Detailed numbers can be found in Table 8.

	AVG	HRL	URL	EN	DE	FR	ES	RU	ZH	JA	TH	TE	BN	SW
NATIVE-EXEMPLARS														
NATIVE-COT	48.1	47.9	44.9	**62.4**	49.2	46.4	56.8	48.4	46.8	40.0	52.8	45.6	46.0	35.2
EN-COT	**51.3**	**52.3**	**46.8**	**62.4**	**53.6**	**51.2**	**58.0**	**55.6**	46.0	**49.6**	**49.6**	**46.8**	46.4	**44.4**
MULTILINGUAL-EXEMPLARS														
NATIVE-COT	29.8	31.8	26.3	52.0	41.6	7.2	10.4	36.0	42.8	32.8	18.0	33.6	26.8	26.8
EN-COT	48.7	50.0	46.3	57.6	53.2	43.2	53.2	48.0	**51.2**	43.6	46.8	46.4	**48.4**	43.6
ENGLISH-EXEMPLARS														
EN-COT	34.7	39.4	26.6	**62.4**	46.0	37.2	50.4	23.6	29.2	26.8	17.2	30.0	34.4	24.8

Table 4: Performance on MGSM with different prompt exemplar type choices: the first section is copied correspondingly from Table 3. The best numbers in each column are in **boldface**.

their presence in training data and that language models are able to transfer their knowledge from high-resource to underrepresented languages to some extent.

Effect of model scale. We analyze the effect of model scale (i.e., number of model parameters and computational resources used for training) on their multilingual arithmetic reasoning abilities (Figure 4). As the models scale up, the performance generally improves for both GPT-3 and PaLM model series on all languages. Neither model achieves a substantial solve rate until a certain scale (`text-davinci-001` for GPT-3 and PaLM-62B for PaLM), hence multilingual reasoning can be considered an *emergent ability* of large language models (Wei et al., 2022a). It is worth noting that the amount of training data per language is constant across language model scales for PaLM—the fact that scale facilitates reasoning implies that further scaling may continue to improve the multilingual reasoning ability of large language models.

Effect of exemplar amount. We analyze how the multilingual reasoning performance of PaLM-540B, the overall best-performing model, is affected by the number of few-shot exemplars (Figure 5). Although not all trends are strictly increasing with the number of exemplars, PaLM-540B benefits from having more examples in general for all languages.